Preprocessing

ACTL3143 & ACTL5111 Deep Learning for Actuaries

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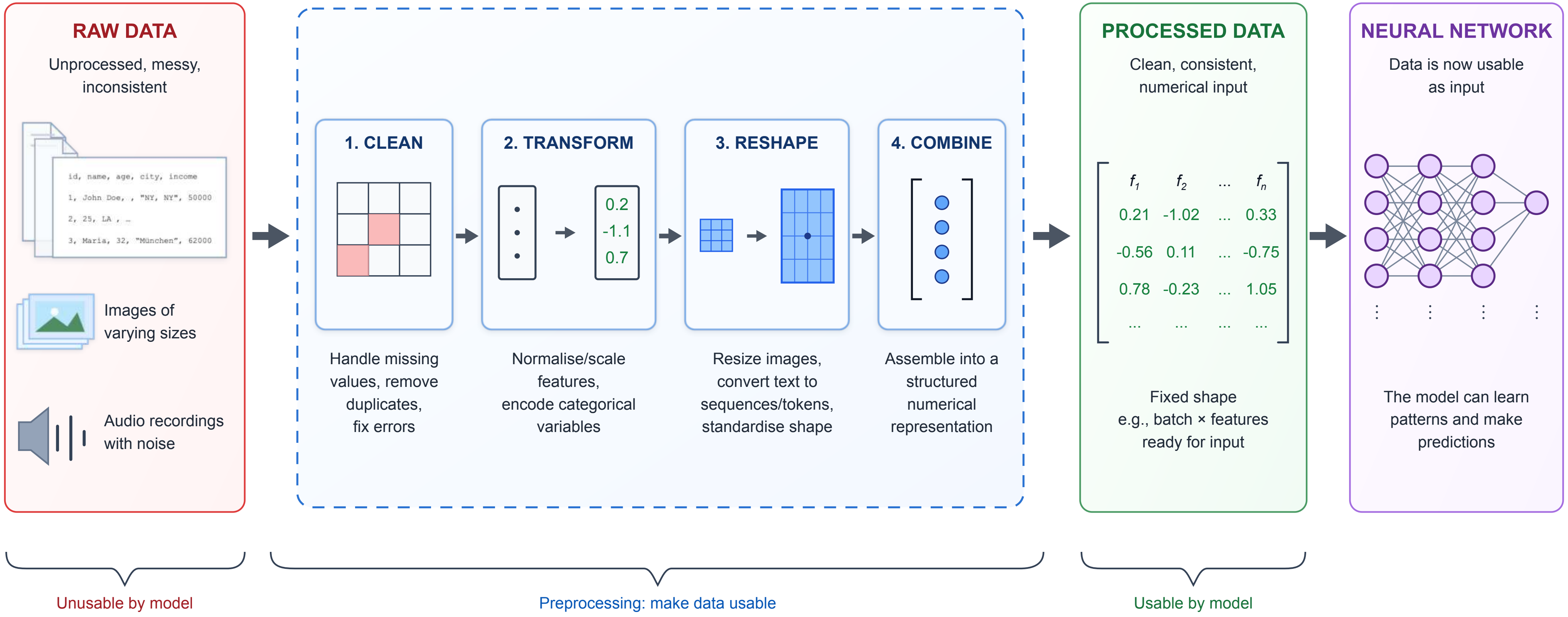
Deep learning project life cycle

1. Define objectives
2. *Collect & label data*
3. Clean & preprocess data (opt. *feature engineering*)
4. Design network architecture
5. Train and validate model
6. Hyperparameter tuning
7. Test and evaluate performance
8. *Deploy to production*
9. *Monitor and iterate*

(We won't cover the italicised items in this course, though they are very important.)

Overview



Preprocessing refers to the transformations of the raw data before input to the network. Some steps are specific to neural networks (e.g. standardising continuous variables), others are necessary for every model (e.g. encoding categorical variables, handling missing data).

Imports needed for these demos

```python
import random
from pathlib import Path

import joblib
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

from keras.models import Sequential
from keras.layers import Dense, Input
from keras.callbacks import EarlyStopping

import sklearn
from sklearn import set_config
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import OneHotEncoder, OrdinalEncoder, StandardScaler
from sklearn.impute import SimpleImputer
from sklearn.compose import make_column_transformer
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LinearRegression
```

Lecture Outline

- **Standardising the Numerical Covariates**
- Handling Categorical Variables
- Missing Values and Rare Observations
- Procedures to Keep Your Data Organised

California housing dataset

```python
features, target = sklearn.datasets.fetch_california_housing(
    as_frame=True, return_X_y=True)
features
```

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Lati
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.8
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.8
...	...	...	...	...	...	...	...
20638	1.8672	18.0	5.329513	1.171920	741.0	2.123209	39.4
20639	2.3886	16.0	5.254717	1.162264	1387.0	2.616981	39.3

20640 rows × 8 columns

Pandas will assign or guess dtypes

Can look at the data types of the features.

```
features.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20640 entries, 0 to 20639
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   MedInc      20640 non-null  float64
 1   HouseAge    20640 non-null  float64
 2   AveRooms    20640 non-null  float64
 3   AveBedrms   20640 non-null  float64
 4   Population  20640 non-null  float64
 5   AveOccup    20640 non-null  float64
 6   Latitude    20640 non-null  float64
 7   Longitude   20640 non-null  float64
dtypes: float64(8)
memory usage: 1.3 MB
```

Here, all the features were continuous variables

Pandas can guess wrong though

For example, Australian postcodes are four digits and some start with `0` (e.g. Darwin is `0800`). A CSV reader just sees digits, guesses a number, and silently drops the leading zero — giving the column the wrong *type* **and** the wrong *value*.

The raw file keeps the leading zero:

```
print(Path("data/postcodes.csv").read_text())
```

```
suburb,postcode
Darwin,0800
Sydney,2000
Perth,6000
```

Default inference:

```
df = pd.read_csv("data/postcodes.csv")
df
```

	suburb	postcode
0	Darwin	800
1	Sydney	2000
2	Perth	6000

```
df.dtypes
```

```
suburb          str
postcode      int64
dtype: object
```

Typed at read time:

```
df = pd.read_csv("data/postcodes.csv",
    dtype={"postcode": "string"})
df
```

	suburb	postcode
0	Darwin	0800
1	Sydney	2000
2	Perth	6000

```
df.dtypes
```

```
suburb           str
postcode      string
dtype: object
```

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Splitting into train, validation and test sets

```
X_main, X_test_raw, y_main, y_test = train_test_split(
    features, target, test_size=0.2, random_state=1
)
X_train_raw, X_val_raw, y_train, y_val = train_test_split(
    X_main, y_main, test_size=0.25, random_state=1
)
```

Note that we really cannot allow there to be duplicate rows in the training, validation, or test sets. If there are, we will have data leakage between the sets, which will lead to overfitting and poor generalisation.

Checking for leaks

We can look for rows that appear in multiple of the datasets.

```
train_val_overlap = pd.merge(X_train_raw, X_val_raw, how='inner')
train_test_overlap = pd.merge(X_train_raw, X_test_raw, how='inner')
val_test_overlap = pd.merge(X_val_raw, X_test_raw, how='inner')

print(f"Number of duplicated rows between train and val: {len(train_val_overlap)}")
print(f"Number of duplicated rows between train and test: {len(train_test_overlap)}")
print(f"Number of duplicated rows between val and test: {len(val_test_overlap)}")
```

```
Number of duplicated rows between train and val: 0
Number of duplicated rows between train and test: 0
Number of duplicated rows between val and test: 0
```

An earlier check is to check that there are no duplicate rows in the dataset.

```
features.duplicated().sum()
```

```
np.int64(0)
```

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We rescaled every column

```python
scaler = StandardScaler()
scaler.fit(X_train_raw)

X_train = scaler.transform(X_train_raw)
X_val = scaler.transform(X_val_raw)
X_test = scaler.transform(X_test_raw)

X_train
```

```
array([[ 0.15, -0.76,  0.26, ..., -0.01, -0.47,  0.69],
       [-1.79, -1.48,  0.48, ..., -0.08, -0.44,  1.33],
       [-0.97, -0.13, -0.67, ...,  0.05, -0.86,  0.65],
       ...,
       [ 0.11, -0.84, -0.54, ..., -0.03, -0.81,  0.6 ],
       [-0.82,  0.99, -0.03, ..., -0.03,  0.54, -0.11],
       [ 0.9 , -1.56,  0.66, ...,  0.07, -0.75,  0.97]], shape=(12384, 8))
```

That is because the neural network expects continuous inputs to be normalised to help with the training procedure.

> **Note**
>
> Notice that the output of `X_train` here looks a bit different to `X_train_raw` earlier? We'll touch on this in a couple of slides.

Scikit-learn preprocessing methods

- `fit`: learn the parameters of the transformation
- `transform`: apply the transformation
- `fit_transform`: learn the parameters and apply the transformation

fit | fit_transform

```
scaler = StandardScaler()
scaler.fit(X_train_raw)
X_train = scaler.transform(X_train_raw)
X_val = scaler.transform(X_val_raw)
X_test = scaler.transform(X_test_raw)

print(X_train.mean(axis=0))
print(X_train.std(axis=0))
print(X_val.mean(axis=0))
print(X_val.std(axis=0))
```

```
[ 1.14e-16 -1.08e-16 -2.82e-16 -4.45e-17 -6.89e-18
-4.02e-17  1.14e-15
  1.75e-15]
[1. 1. 1. 1. 1. 1. 1. 1.]
[ 0.01  0.01  0.   -0.01 -0.    0.01  0.   -0.01]
[1.01 1.   0.81 0.81 0.98 0.84 0.99 1.  ]
```

Dataframes & arrays

```
1  X_test_raw
```

	MedInc	HouseAge	AveRooms
4712	3.2500	39.0	4.503205
2151	1.9784	37.0	4.988584
...	...	...	...
6823	4.8750	42.0	5.347985
11878	2.7054	52.0	5.741214

4128 rows × 8 columns

```
1  X_test
```

```
array([[-0.33,  0.83, -0.34, ..., -0.11,
-0.73,  0.6 ],
       [-1.  ,  0.67, -0.16, ..., -0.04,
0.54, -0.1 ],
       [ 0.07,  1.38, -0.35, ...,  0.06,
0.98, -1.42],
        ...,
       [ 0.62, -0.21, -0.16, ...,  0.05,
0.92, -1.4 ],
       [ 0.53,  1.07, -0.03, ..., -0.  ,
-0.72,  0.73],
       [-0.62,  1.86,  0.11, ..., -0.02,
-0.77,  1.09]], shape=(4128, 8))
```

Note

By default, when you pass `sklearn` a DataFrame it returns a `numpy` array.

Keep as a DataFrame

From scikit-learn 1.2:

```
set_config(transform_output="pandas")

imp = SimpleImputer()

X_train = imp.fit_transform(X_train_raw)
X_val = imp.transform(X_val_raw)
X_test = imp.transform(X_test_raw)
```

```
X_test
```

	MedInc	HouseAge	AveRoo
4712	3.2500	39.0	4.503205
2151	1.9784	37.0	4.988584
...	...	...	...
6823	4.8750	42.0	5.347985
11878	2.7054	52.0	5.741214

4128 rows × 8 columns

From SimpleImputer's docs:

> Replace missing values using a descriptive statistic (e.g. mean, median, or most frequent) along each column, or using a constant value… default='mean'

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French motor dataset

```python
 # Download the dataset if we don't have it already.
if not Path("data/freq_data.csv").exists():
    freq = sklearn.datasets.fetch_openml(data_id=41214, as_frame=True).frame
    freq.to_csv("data/freq_data.csv", index=False)
else:
    freq = pd.read_csv("data/freq_data.csv")
freq.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 678013 entries, 0 to 678012
Data columns (total 12 columns):
 #   Column      Non-Null Count   Dtype
---  ------      --------------   -----
 0   IDpol       678013 non-null  float64
 1   ClaimNb     678013 non-null  float64
 2   Exposure    678013 non-null  float64
 3   Area        678013 non-null  str
 4   VehPower    678013 non-null  float64
 5   VehAge      678013 non-null  float64
 6   DrivAge     678013 non-null  float64
 7   BonusMalus  678013 non-null  float64
 8   VehBrand    678013 non-null  str
 9   VehGas      678013 non-null  str
 10  Density     678013 non-null  float64
```

The data

```
freq
```

	IDpol	ClaimNb	Exposure	Area	VehPower	VehAge	DrivAge	Bon
0	1.0	1.0	0.10000	D	5.0	0.0	55.0	50.0
1	3.0	1.0	0.77000	D	5.0	0.0	55.0	50.0
2	5.0	1.0	0.75000	B	6.0	2.0	52.0	50.0
...	...	...	...	...	...	...	...	...
678010	6114328.0	0.0	0.00274	D	6.0	2.0	45.0	50.0
678011	6114329.0	0.0	0.00274	B	4.0	0.0	60.0	50.0
678012	6114330.0	0.0	0.00274	B	7.0	6.0	29.0	54.0

678013 rows × 12 columns

Data dictionary

- `IDpol`: policy number (unique identifier)
- `ClaimNb`: number of claims on the given policy
- `Exposure`: total exposure in yearly units
- `Area`: area code (categorical, ordinal)
- `VehPower`: power of the car (categorical, ordinal)
- `VehAge`: age of the car in years
- `DrivAge`: age of the (most common) driver in years
- `BonusMalus`: bonus-malus level between 50 and 230 (with reference level 100)
- `VehBrand`: car brand (categorical, nominal)
- `VehGas`: diesel or regular fuel car (binary)
- `Density`: of inhabitants per km^2 in the city of the living place of the driver
- `Region`: regions in France (prior to 2016)

We have $\{(\mathbf{x}_i, y_i)\}_{i=1,\dots,n}$ for $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \mathbb{N}_0$. Assume the distribution

$$Y_i \sim \mathsf{Poisson}(\lambda(\mathbf{x}_i))$$

We have $\mathbb{E}Y_i = \lambda(\mathbf{x}_i)$. The NN takes $\mathbf{x}_i$ & predicts $\mathbb{E}Y_i$.

```python
freq = freq.drop("IDpol", axis=1)
```


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Split the data

```
X_train_raw, X_test_raw, y_train, y_test = train_test_split(
  freq.drop("ClaimNb", axis=1), freq["ClaimNb"], random_state=2023)
X_train_raw = X_train_raw.reset_index(drop=True)
X_test_raw = X_test_raw.reset_index(drop=True)
X_train_raw
```

	Exposure	Area	VehPower	VehAge	DrivAge	BonusMalus	VehBrand
0	0.01	A	4.0	0.0	70.0	50.0	B12
1	1.00	C	6.0	11.0	36.0	50.0	B3
2	0.08	D	7.0	9.0	35.0	50.0	B2
...	...	...	...	...	...	...	...
508506	0.22	E	7.0	21.0	32.0	90.0	B5
508507	1.00	C	7.0	15.0	51.0	50.0	B1
508508	0.42	D	6.0	1.0	37.0	54.0	B2

508509 rows × 10 columns

What values do we see in the data?

```python
X_train_raw["Area"].value_counts()
X_train_raw["VehBrand"].value_counts()
X_train_raw["VehGas"].value_counts()
X_train_raw["Region"].value_counts()
```

```
Area
C    144156
D    113532
E    102791
A     78017
B     56571
F     13442
Name: count, dtype: int64
```

```
VehBrand
B12     124490
B1      122200
B2      120155
         ...
B11      10218
B13       9160
B14       2998
Name: count, Length: 11, dtype: int64
```

```
VehGas
Regular    259422
Diesel     249087
Name: count, dtype: int64
```

```
Region
R24     120597
R82      63555
R93      59627
         ...
R21       2283
R42       1634
R43       1024
Name: count, Length: 22, dtype: int64
```

Nominal categorical variables

These are categorical variables which you cannot order in a 'default' sensible way.

```python
gas = X_train_raw[["VehGas"]]
```

The original data:

```python
gas.head()
```

	VehGas
0	Regular
1	Regular
2	Diesel
3	Regular
4	Diesel

One-hot encoding

```python
enc = OneHotEncoder(sparse_output=False)

X_train = enc.fit_transform(gas)
X_train.head()
```

	VehGas_Diesel	VehGas_Regular
0	0.0	1.0
1	0.0	1.0
2	1.0	0.0
3	0.0	1.0
4	1.0	0.0

Dummy encoding

```python
enc = OneHotEncoder(
    drop="first",
    sparse_output=False)
X_train = enc.fit_transform(gas)
X_train.head()
```

	VehGas_Regular
0	1.0
1	1.0
2	0.0
3	1.0
4	0.0

⚠ **Warning**

Collinearity is not a big deal for neural networks. However if you want to reuse the same `X_train` data for a (generalised) linear regression, then you probably want to dummy-encode here to avoid needing two different preprocessing regimes for the two competing models.

Ordinal categorical variables

These are categorical variables which have a natural order to them.

```
enc = OrdinalEncoder()
enc.fit(X_train_raw[["Area"]])
enc.categories_
```

```
[array(['A', 'B', 'C', 'D', 'E', 'F'], dtype=object)]
```

```
for i, area in enumerate(enc.categories_[0]):
    print(f"The Area value {area} gets turned into {i}.")
```

```
The Area value A gets turned into 0.
The Area value B gets turned into 1.
The Area value C gets turned into 2.
The Area value D gets turned into 3.
The Area value E gets turned into 4.
The Area value F gets turned into 5.
```

```
print(f"In other words, {' < '.join(enc.categories_[0])}, and ")
print(" = ".join(f"{r} - {l}" for l, r in zip(enc.categories_[0][:-1], enc.categories_[0]
```

```
In other words, A < B < C < D < E < F, and
B - A = C - B = D - C = E - D = F - E
```

Ordinal encoded values

```python
X_train = enc.transform(X_train_raw[["Area"]])
X_test = enc.transform(X_test_raw[["Area"]])
```

```python
X_train_raw[["Area"]].head()
```

	Area
0	A
1	C
2	D
3	E
4	B

```python
X_train.head()
```

	Area
0	0.0
1	2.0
2	3.0
3	4.0
4	1.0

We could train on this `X_train`, or on the previous `X_train`, but each only contains one feature from the dataset. How do we add the continuous variables back in? Use a sklearn *column transformer* for that.

Transform all columns in one hit I

```python
ct = make_column_transformer(
  (OneHotEncoder(sparse_output=False, drop="first"), ["VehGas", "VehBrand", "Region"]),
  (OrdinalEncoder(), ["Area"]),
  remainder=StandardScaler()
)

X_train = ct.fit_transform(X_train_raw)
```

```python
X_train_raw
```

	Exposure	Area	VehPower	V
0	0.01	A	4.0	0.
...	...	...	...	...
508508	0.42	D	6.0	1.

508509 rows × 10 columns

```python
X_train
```

	onehotencoder__VehGas_Regul
0	1.0
...	...
508508	0.0

508509 rows × 39 columns

Transform all columns in one hit II

```python
ct = make_column_transformer(
  (OneHotEncoder(sparse_output=False, drop="first"), ["VehGas", "VehBrand", "Region"]),
  (OrdinalEncoder(), ["Area"]),
  remainder=StandardScaler(),
  verbose_feature_names_out=False
)
X_train = ct.fit_transform(X_train_raw)
```

```python
X_train_raw
```

	Exposure	Area	VehPower	V
0	0.01	A	4.0	0.
...	...	...	...	...
508508	0.42	D	6.0	1.

508509 rows × 10 columns

```python
X_train
```

	VehGas_Regular	VehBrand_B1
0	1.0	0.0
...	...	...
508508	0.0	0.0

508509 rows × 39 columns

Aside: The order of the variables matters

```python
X_train_raw["Area_In_Words"] = X_train_raw["Area"].map({
    "A": "Very low", "B": "Low", "C": "Medium",
    "D": "High", "E": "Very high", "F": "Extremely high"
})
```

```python
enc_wrong = OrdinalEncoder()
enc_wrong.fit(X_train_raw[["Area_In_Words"]])
enc_wrong.categories_
```

```
[array(['Extremely high', 'High', 'Low', 'Medium', 'Very high', 'Very low'],
       dtype=object)]
```

```python
for i, area in enumerate(enc_wrong.categories_[0]):
    print(f"The Area value {area} gets turned into {i}.")
```

```
The Area value Extremely high gets turned into 0.
The Area value High gets turned into 1.
The Area value Low gets turned into 2.
The Area value Medium gets turned into 3.
The Area value Very high gets turned into 4.
The Area value Very low gets turned into 5.
```

```python
print(f"In other words, {' < '.join(enc_wrong.categories_[0])}.")
```

```
In other words, Extremely high < High < Low < Medium < Very high < Very low.
```

Aside: Manually set the order

```python
ordered = [["Very low", "Low", "Medium", "High", "Very high", "Extremely high"]]
enc_right = OrdinalEncoder(categories=ordered)
enc_right.fit(X_train_raw[["Area_In_Words"]])
enc_right.categories_
```

```
[array(['Very low', 'Low', 'Medium', 'High', 'Very high', 'Extremely high'],
       dtype=object)]
```

```python
for i, area in enumerate(enc_right.categories_[0]):
    print(f"The Area value {area} gets turned into {i}.")
```

```
The Area value Very low gets turned into 0.
The Area value Low gets turned into 1.
The Area value Medium gets turned into 2.
The Area value High gets turned into 3.
The Area value Very high gets turned into 4.
The Area value Extremely high gets turned into 5.
```

```python
print(f"In other words, {' < '.join(enc_right.categories_[0])}.")
```

```
In other words, Very low < Low < Medium < High < Very high < Extremely high.
```

```python
X_train_raw = X_train_raw.drop("Area_In_Words", axis=1)
```

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Stroke prediction dataset

Dataset source: Kaggle Stroke Prediction Dataset.

```python
RAW_DATA_DIR = Path("stroke/raw")
data = pd.read_csv(RAW_DATA_DIR / "stroke.csv")
data.head()
```

	id	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level
0	9046	Male	67.0	0	1	Yes	Private	Urban	228.69
1	51676	Female	61.0	0	0	Yes	Self-employed	Rural	202.21
2	31112	Male	80.0	0	1	Yes	Private	Rural	105.92
3	60182	Female	49.0	0	0	Yes	Private	Urban	171.23
4	1665	Female	79.0	1	0	Yes	Self-employed	Rural	174.12

Data description

1. `id`: unique identifier
2. `gender`: "Male", "Female" or "Other"
3. `age`: age of the patient
4. `hypertension`: 0 or 1 if the patient has hypertension
5. `heart_disease`: 0 or 1 if the patient has any heart disease
6. `ever_married`: "No" or "Yes"
7. `work_type`: "children", "Govt_job", "Never_worked", "Private" or "Self-employed"
8. `Residence_type`: "Rural" or "Urban"
9. `avg_glucose_level`: average glucose level in blood
10. `bmi`: body mass index
11. `smoking_status`: "formerly smoked", "never smoked", "smokes" or "Unknown"
12. `stroke`: 0 or 1 if the patient had a stroke

Source: Kaggle, Stroke Prediction Dataset.

Look for missing values and split the data

First, look for missing values.

```
number_missing = data.isna().sum()
number_missing[number_missing > 0]
```

```
bmi     201
dtype: int64
```

Tip

Always take a look at the missing data in, say, Excel. Sometimes pandas reads in a category like “None” and interprets that as a missing value, when it’s really a valid category. This happened to me just the other day.

```
features = data.drop(["id", "stroke"], axis=1)
target = data["stroke"]

X_main_raw, X_test_raw, y_main, y_test = train_test_split(
    features, target, test_size=0.2, random_state=7)
X_train_raw, X_val_raw, y_train, y_val = train_test_split(
    X_main_raw, y_main, test_size=0.25, random_state=12)

X_train_raw.shape, X_val_raw.shape, X_test_raw.shape
```

```
((3066, 10), (1022, 10), (1022, 10))
```

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What values do we see in the data?

```
X_train_raw["gender"].value_counts()
```

```
gender
Female    1802
Male      1264
Name: count, dtype: int64
```

```
X_train_raw["ever_married"].value_counts(
```

```
ever_married
Yes    2007
No     1059
Name: count, dtype: int64
```

```
X_train_raw["Residence_type"].value_count
```

```
Residence_type
Urban    1536
Rural    1530
Name: count, dtype: int64
```

```
X_train_raw["work_type"].value_counts()
```

```
work_type
Private          1754
Self-employed     490
children          419
Govt_job          390
Never_worked       13
Name: count, dtype: int64
```

```
X_train_raw["smoking_status"].value_count
```

```
smoking_status
never smoked       1130
Unknown             944
formerly smoked     522
smokes              470
Name: count, dtype: int64
```

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Group sparse categories

Look for sparse categories. One common trick is to lump together all the small categories (e.g. < 5% or 10% each) into a new combined 'OTHER' or 'RARE' category. The `OneHotEncoder` can group infrequent levels for us.

```
X_train_raw["work_type"].value_counts()
```

```
work_type
Private          1754
Self-employed     490
children          419
Govt_job          390
Never_worked       13
Name: count, dtype: int64
```

min_frequency | max_categories

Lump together any level below 5% of the rows:

```
enc = OneHotEncoder(sparse_output=False,
    min_frequency=0.05)
ohe = enc.fit_transform(X_train_raw[["work_type"]])
enc.infrequent_categories_
```

```
[array(['Never_worked'], dtype=object)]
```

```
ohe.head(3)
```

	work_type_Govt_job	work_type_Private	work_type_Self-employed	work_ty
1914	0.0	1.0	0.0	0.0
1958	0.0	0.0	0.0	1.0
2920	0.0	1.0	0.0	0.0

Make a plan for preprocessing each column

1. Take categorical columns $\hookrightarrow$ one-hot vectors
2. binary columns $\hookrightarrow$ do nothing (already 0 & 1)
3. continuous columns $\hookrightarrow$ impute NaNs & standardise.

Make the column transformer

```python
nom_vars =  ["gender", "ever_married", "Residence_type",
    "work_type", "smoking_status"]

ct = make_column_transformer(
  (OneHotEncoder(sparse_output=False, handle_unknown="ignore"), nom_vars),
  ("passthrough", ["hypertension", "heart_disease"]),
  remainder=make_pipeline(SimpleImputer(), StandardScaler()),
  verbose_feature_names_out=False
)

X_train = ct.fit_transform(X_train_raw)
X_val = ct.transform(X_val_raw)
X_test = ct.transform(X_test_raw)

for name, X in zip(("train", "val", "test"), (X_train, X_val, X_test)):
    num_na = pd.DataFrame(X).isna().sum().sum()
    print(f"The {name} set has shape {X.shape} & with {num_na} NAs.")
```

```
The train set has shape (3066, 20) & with 0 NAs.
The val set has shape (1022, 20) & with 0 NAs.
The test set has shape (1022, 20) & with 0 NAs.
```

Handling unseen categories

```
X_train_raw["gender"].value_counts()
```

```
gender
Female    1802
Male      1264
Name: count, dtype: int64
```

```
X_val_raw["gender"].value_counts()
```

```
gender
Female    615
Male      406
Other       1
Name: count, dtype: int64
```

```
ind = np.argmax(X_val_raw["gender"] == "O
X_val_raw.iloc[ind-1:ind+3][["gender"]]
```

	gender
4970	Male
3116	Other
4140	Male
2505	Female

```
gender_cols = pd.DataFrame(X_val)[["gende
gender_cols.iloc[ind-1:ind+3]
```

	gender_Female	gender_Male
4970	0.0	1.0
3116	0.0	0.0
4140	0.0	1.0
2505	1.0	0.0

Handling unseen categories II

This was caused by fitting the encoder on the `X_train` which, by bad luck, didn't have the "Other" category. Why not just read the entire dataset before splitting to collect all the values the categorical variables could take?

It's a very mild form of data leakage, though it may be justifiable in some cases. If we made the datasets, we'd have a better idea of whether it makes sense on a case-by-case basis. We struggle here as we (in this course) are just analysing datasets which we didn't collect.

Lecture Outline

- Standardising the Numerical Covariates
- Handling Categorical Variables
- Missing Values and Rare Observations
- **Procedures to Keep Your Data Organised**

Directory structure

A popular convention (from Cookiecutter Data Science) splits `data/` by *how far along* the data is:

```
data/
├── raw/        # original, never edited
├── interim/    # cleaned, still readable
└── processed/  # final, model-ready
```

- **`raw/`** is *immutable* — never overwrite it, so the whole pipeline can always be rebuilt from scratch.
- **`interim/`** is data that can be recreated from raw & a preprocessing script.
- **`processed/`** is what the model loads: split, encoded and scaled.

`interim/` holds data that is **nearly model-ready but still human-readable** — after cleaning, joining and splitting, but *before* rescaling strips the units and one-hot encoding turns everything into a wall of 0s and 1s.

- You can **open it and eyeball it**: ages still look like ages, premiums still in dollars.
- It **caches** slow steps (parsing, joining, geocoding) so tweaking the final features doesn't re-run them.
- It keeps `processed/` **unambiguous** — one obvious file per train/validation/test split.

Save the preprocessed data to files

```
PROCESSED_DATA_DIR = Path("stroke/processed")
PROCESSED_DATA_DIR.mkdir(parents=True, exist_ok=True)

X_train.to_csv(PROCESSED_DATA_DIR / "x_train.csv", index=False)
X_val.to_csv(PROCESSED_DATA_DIR / "x_val.csv", index=False)
X_test.to_csv(PROCESSED_DATA_DIR / "x_test.csv", index=False)
y_train.to_csv(PROCESSED_DATA_DIR / "y_train.csv", index=False)
y_val.to_csv(PROCESSED_DATA_DIR / "y_val.csv", index=False)
y_test.to_csv(PROCESSED_DATA_DIR / "y_test.csv", index=False)
```

Keep the preprocessor around

Later want to make a prediction on some new data point. It has to go through the **exact same** preprocessing steps.

```
joblib.dump(ct, PROCESSED_DATA_DIR / "preprocessor.pkl")
preprocessor = joblib.load(PROCESSED_DATA_DIR / "preprocessor.pkl")
```

```
X_test_raw.head(1)
```

	gender	age	hypertension	heart_disease	ever_married	work_type	Residence_type	avg_glucose_level	bı
2804	Female	69.0	0	0	Yes	Govt_job	Rural	70.98	3C

```
new_data = X_test_raw.head(1).copy()
new_data["ever_married"] = "No" # Let's pretend this is the new data point
```

```
ct.transform(new_data)
```

	gender_Female	gender_Male	ever_married_No	ever_married_Yes	Residence_type_Rural	Residence_type_I
2804	1.0	0.0	1.0	0.0	1.0	0.0

```
preprocessor.transform(new_data)
```

	gender_Female	gender_Male	ever_married_No	ever_married_Yes	Residence_type_Rural	Residence_type_I
2804	1.0	0.0	1.0	0.0	1.0	0.0

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Glossary

- column transformer
- nominal variables
- ordinal variables
- one-hot encoding
- missing values
- imputation
- raw / interim / processed data
- standardisation